

Partial Results on the Convexity Decision Problem for Coordinate Projections of $\text{SO}(n)$

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Abstract

Ramachandran, Shu, and Wang asked whether the following decision problem is NP-hard: given a coordinate set $S \subseteq [n] \times [n]$, decide whether the coordinate projection $\pi_S(\text{SO}(n))$ is convex. This note does not resolve that conjecture. It records partial progress and several rigorous checks around a natural attempted reduction.

The main verified points are as follows. First, under the standard dense support-matrix encoding, the convexity problem is reducible to the universal-existential theory of the reals; in particular it lies in $\forall\exists\mathbb{R} \subseteq \text{PSPACE}$. Second, the natural sufficient condition that S be contained, up to row and column permutations and transposition, in the strict upper triangular pattern is NP-complete to recognize. Third, this NP-completeness result is not an NP-hardness proof for the convexity problem: strict triangularizability is sufficient but not necessary for convexity. Finally, we give elementary hereditary nonconvexity certificates and an explicit nonconvex 4×4 pattern that avoids those simple certificates. The note ends by isolating the missing gadget statement that would be needed to turn the triangularizability hardness into a proof of NP-hardness for the original problem.

1 Status and problem statement

For $n \geq 2$ let

$$\text{SO}(n) = \{X \in \mathbb{R}^{n \times n} : X^T X = I, \det X = 1\}.$$

For a coordinate set $S \subseteq [n] \times [n]$, let

$$P_S(n) := \pi_S(\text{SO}(n)) \subseteq \mathbb{R}^S,$$

where π_S keeps exactly the entries indexed by S . The decision problem considered by Ramachandran, Shu, and Wang is

$$\text{CONVEX} : \quad (n, S) \mapsto \begin{cases} \text{TRUE}, & P_S(n) \text{ is convex,} \\ \text{FALSE}, & P_S(n) \text{ is not convex.} \end{cases}$$

Their Conjecture 1 states that CONVEX is NP-hard [7]. The same paper proves, among other results, that every two-dimensional linear image of $\text{SO}(n)$ is convex and that the projection of $\text{SO}(n)$ onto the strict upper triangular entries is convex [7].

Status of this note. This note gives partial progress only. It does not prove NP-hardness of CONVEX, nor does it give a characterization of the coordinate sets S for which $P_S(n)$ is convex. The main point is to separate the part of a tempting reduction that is correct from the part that remains missing.

Remark 1.1 (Encoding convention). The complexity statements below use the standard dense encoding in which the input contains the $n \times n$ Boolean incidence matrix of S , so the input length is at least on the order of n^2 . This is the natural encoding for the coordinate-pattern problem and is the one relevant to the reductions discussed here. If n is written in binary and S is given sparsely, the formula in Theorem 3.1 is still polynomial in n and $|S|$, but not necessarily polynomial in the bit length of that sparse input.

2 Symmetries and heredity

We shall use the following elementary invariances repeatedly.

Lemma 2.1 (Coordinate-pattern symmetries). *Convexity of $P_S(n)$ is invariant under independent row permutations, independent column permutations, and transposition of the pattern S .*

Proof. Transposition is induced by $X \mapsto X^T$, which preserves $\text{SO}(n)$. For row and column permutations, let P, Q be permutation matrices. If $\det(PQ) = 1$, then $X \mapsto PXQ$ is a bijection of $\text{SO}(n)$ and induces the corresponding permutation of coordinates. If $\det(PQ) = -1$, multiply one of P, Q by a diagonal sign matrix of determinant -1 . This restores determinant 1 and changes the target coordinates only by coordinate sign flips. Since coordinate sign flips are invertible linear maps of $\mathbb{R}^S$, they preserve convexity and nonconvexity. $\square$

Remark 2.2. The preceding proof only uses sign changes induced by determinant-correcting row or column sign matrices. It does not assert that arbitrary independent sign changes on individual selected coordinates are themselves symmetries of the support pattern.

Lemma 2.3 (Heredity). *If $T \subseteq S \subseteq [n] \times [n]$ and $P_S(n)$ is convex, then $P_T(n)$ is convex. Equivalently, if $P_T(n)$ is nonconvex for some $T \subseteq S$, then $P_S(n)$ is nonconvex.*

Proof. The set $P_T(n)$ is obtained from $P_S(n)$ by a linear coordinate projection. Linear images of convex sets are convex. $\square$

3 A universal-existential upper bound

Let $\forall\exists\mathbb{R}$ denote the class of decision problems polynomial-time reducible to the truth problem for sentences of the form

$$\forall u \in \mathbb{R}^a \exists v \in \mathbb{R}^b \quad \Phi(u, v),$$

where Φ is a quantifier-free Boolean combination of polynomial equalities and inequalities with rational coefficients. It is known that

$$\forall\exists\mathbb{R} \subseteq \text{PSPACE};$$

see, for example, [3, 5], building on real-quantifier-elimination and real-feasibility algorithms such as [2, 6].

Theorem 3.1 (Upper bound). *With the dense support-matrix encoding, CONVEX belongs to $\forall\exists\mathbb{R}$. Consequently,*

$$\text{CONVEX} \in \text{PSPACE}.$$

Proof. Convexity of $P_S(n)$ is equivalent to the following assertion: for all $X, Y \in \text{SO}(n)$ and all $\lambda \in [0, 1]$, there exists $Z \in \text{SO}(n)$ such that

$$Z_{ij} = \lambda X_{ij} + (1 - \lambda)Y_{ij} \quad ((i, j) \in S).$$

This is already a universal-existential statement over the reals, except that one must encode the determinant constraints with polynomial-size syntax.

Fix a polynomial-size arithmetic circuit C_n over $\mathbb{Z}$ computing the determinant of an $n \times n$ matrix. Such circuits are standard; for instance, one may use Berkowitz's determinant construction [1]. For a matrix variable X and a vector U of gate variables, let

$$\text{DetCirc}_n(X, U)$$

be the conjunction of the equations encoding the gates of C_n , and let U_{out} be the output gate. Every gate equation has degree at most two, and the number of gate variables and equations is polynomial in n .

Let

$$\text{Orth}(X) := \bigwedge_{1 \leq a, b \leq n} \sum_{k=1}^n X_{ka} X_{kb} = \delta_{ab}.$$

Consider the sentence

$$\begin{aligned} \forall X, Y, \lambda, U, V \exists Z, W \quad & \left(\text{Orth}(X) \wedge \text{DetCirc}_n(X, U) \wedge U_{\text{out}} = 1 \right. \\ & \left. \wedge \text{Orth}(Y) \wedge \text{DetCirc}_n(Y, V) \wedge V_{\text{out}} = 1 \wedge 0 \leq \lambda \leq 1 \right) \\ \implies & \left(\text{Orth}(Z) \wedge \text{DetCirc}_n(Z, W) \wedge W_{\text{out}} = 1 \right. \\ & \left. \wedge \bigwedge_{(i,j) \in S} Z_{ij} = \lambda X_{ij} + (1 - \lambda)Y_{ij} \right). \end{aligned} \quad (*)$$

Here the matrix variables range over $\mathbb{R}^{n \times n}$ and the circuit variables range over real gate assignments. Because the determinant circuit equations uniquely determine their gate values, the universal variables U, V cause no logical difficulty: for the correct gate assignments the antecedent says precisely that $X, Y \in \text{SO}(n)$, and for incorrect gate assignments the antecedent is false.

Thus $(*)$ is true if and only if $P_S(n)$ is convex. Its length is polynomial in the dense input size, since $|S| \leq n^2$ and the determinant circuits have polynomial size. Therefore the map $(n, S) \mapsto (*)$ is a polynomial-time reduction to the universal-existential theory of the reals. The inclusion in PSPACE follows from $\forall \exists \mathbb{R} \subseteq \text{PSPACE}$. $\square$

Remark 3.2. This upper bound does not place CONVEX in NP. It also does not supply an NP-hardness proof. Its role is only to locate the problem within a standard real-algebraic decision class.

4 A sufficient condition and elementary nonconvexity certificates

Definition 4.1 (Strict triangularizability). A pattern $S \subseteq [n] \times [n]$ is *strictly triangularizable* if, after independent row and column permutations, every element of S lies strictly above the diagonal:

$$S \subseteq \{(i, j) : 1 \leq i < j \leq n\}.$$

It is strictly triangularizable up to transposition if either S or S^T has this property.

Lemma 4.2 (Strict triangular sufficient condition). *If S is strictly triangularizable up to transposition, then $P_S(n)$ is convex.*

Proof. Ramachandran, Shu, and Wang prove that the full strict upper triangular projection of $\text{SO}(n)$ is convex [7]. The result follows from Theorems 2.1 and 2.3. $\square$

The next two lemmas give small hereditary certificates of nonconvexity.

Lemma 4.3 (Perfect matching plus an extra edge). *Let $n \geq 3$. Suppose $S \subseteq [n] \times [n]$ contains a perfect matching and at least one additional edge not in that matching. Then $P_S(n)$ is nonconvex.*

Proof. By Theorems 2.1 and 2.3, it is enough to prove the claim for

$$T = \{(1, 1), (2, 2), \dots, (n, n), (1, 2)\}.$$

Indeed, after sending the perfect matching to the diagonal, a simultaneous permutation of rows and columns sends the extra off-diagonal edge to $(1, 2)$.

Let

$$X^+ = I_n, \quad X^- = \text{diag}(-1, -1, 1, \dots, 1).$$

Both lie in $\text{SO}(n)$. The midpoint of their T -projections has

$$x_{11} = 0, \quad x_{22} = 0, \quad x_{12} = 0, \quad x_{kk} = 1 \quad (3 \leq k \leq n).$$

If this midpoint had a lift $Z \in \text{SO}(n)$, then $Z_{kk} = 1$ for every $k \geq 3$. Since the columns and rows of Z have norm one, this forces $Ze_k = e_k$ for $k \geq 3$. Hence the upper-left 2×2 block of Z is an element of $\text{SO}(2)$, so it has the form

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

The equations $Z_{11} = 0$ and $Z_{12} = 0$ would require $\cos \theta = \sin \theta = 0$, impossible. Thus the midpoint is not in $P_T(n)$, so $P_T(n)$, and hence $P_S(n)$, is nonconvex. $\square$

Lemma 4.4 (Pinned two-plane obstruction). *Let $n \geq 3$. Suppose that, after row and column permutations, S contains*

$$\{(1, 1), (1, 2), (3, 3), (4, 4), \dots, (n, n)\}.$$

Then $P_S(n)$ is nonconvex.

Proof. Use the same two matrices

$$X^+ = I_n, \quad X^- = \text{diag}(-1, -1, 1, \dots, 1).$$

The midpoint of their projections to the displayed coordinates has

$$x_{11} = x_{12} = 0, \quad x_{kk} = 1 \quad (3 \leq k \leq n).$$

Any lift $Z \in \text{SO}(n)$ must again fix e_k for $k \geq 3$ and therefore restrict to a 2×2 rotation on the first two coordinates. A 2×2 rotation cannot have both first-row entries equal to zero. Hence the displayed coordinate projection is nonconvex, and Theorem 2.3 proves the claim for S . $\square$

Lemma 4.5 (Large rectangle obstruction). *If S contains a full rectangle $A \times B$ with $A, B \subseteq [n]$ and*

$$|A| + |B| > n,$$

then $P_S(n)$ is nonconvex.

Proof. Ramachandran, Shu, and Wang use the known characterization of rectangular submatrix projections of $\text{SO}(n)$: the full $A \times B$ rectangular projection is convex precisely in the range $|A| + |B| \leq n$ and is nonconvex when $|A| + |B| > n$ [7, Section 8]. Therefore $P_{A \times B}(n)$ is nonconvex. The conclusion for S follows from Theorem 2.3. $\square$

5 Convex patterns not explained by strict triangularity

The strict triangular sufficient condition is not a characterization. This is the point at which the direct triangularizability reduction fails.

5.1 The diagonal pattern

Let

$$D_n = \{(i, i) : 1 \leq i \leq n\}.$$

The diagonal projection $P_{D_n}(n)$ is convex; it is the even parity polytope

$$\text{conv}\{s \in \{\pm 1\}^n : \prod_{i=1}^n s_i = 1\},$$

as recalled in [7]. But D_n is not strictly triangularizable. Indeed, any strictly upper triangular pattern contains no perfect matching: if a perfect matching had all its edges in positions $i < \sigma(i)$, summing over i would give

$$\sum_i i < \sum_i \sigma(i) = \sum_i i,$$

which is impossible. Since D_n itself is a perfect matching, it cannot be strictly triangularized.

5.2 A four-dimensional non-diagonal example

Let

$$F = \{(1, 1), (1, 2), (2, 3), (2, 4)\} \subseteq [4] \times [4].$$

Then $P_F(4)$ is convex, but F is not strictly triangularizable.

First, F is not strictly triangularizable because every column of F contains an edge, whereas any subset of the strict upper triangular pattern in 4×4 has an empty first column after triangularization.

Now identify a point of $\mathbb{R}^F$ with a pair $(a, b) \in \mathbb{R}^2 \times \mathbb{R}^2$, where

$$a = (X_{11}, X_{12}), \quad b = (X_{23}, X_{24}).$$

Necessarily $\|a\|_2 \leq 1$ and $\|b\|_2 \leq 1$. Conversely, suppose $a, b \in \mathbb{R}^2$ satisfy these two inequalities. Choose $p, q \in \mathbb{R}^2$ such that

$$\|p\|_2^2 = 1 - \|a\|_2^2, \quad \|q\|_2^2 = 1 - \|b\|_2^2, \quad p \perp b, \quad q \perp a.$$

This is possible even when $a = 0$ or $b = 0$, since then the corresponding orthogonality condition is vacuous. Define

$$u = (a, p), \quad v = (q, b) \quad \text{in } \mathbb{R}^4.$$

Then $\|u\| = \|v\| = 1$ and

$$\langle u, v \rangle = \langle a, q \rangle + \langle p, b \rangle = 0.$$

Extend u, v to a positively oriented orthonormal basis of $\mathbb{R}^4$; taking these vectors as the first two rows gives a matrix in $\text{SO}(4)$ with the prescribed F -coordinates. Hence

$$P_F(4) = B_2 \times B_2,$$

which is convex.

This example agrees with the 4×4 classification in [7, Section 8], where, up to symmetries, every convex coordinate pattern is contained in the strict upper triangular pattern, the diagonal pattern, or the pattern F above.

Corollary 5.1. *Strict triangularizability is a sufficient but not necessary condition for convexity of $P_S(n)$.*

6 Strict triangularizability is NP-complete

Definition 6.1. A square 0/1 pattern $E \subseteq [m] \times [m]$ is *weakly triangularizable* if its rows and columns can be independently permuted so that every edge lies in a weakly upper triangular position $i \leq j$.

Fertin, Rusu, and Vialette proved that weak triangularizability of square 0/1 matrices is NP-complete [4]. The strict version follows by a short padding reduction.

Proposition 6.2 (Strict triangularizability is NP-complete). *Given m and $E \subseteq [m] \times [m]$, deciding whether E is strictly triangularizable is NP-complete.*

Proof. Membership in NP is immediate: the certificate consists of the two permutations, and the strict inequalities can be checked in polynomial time.

For hardness, reduce from weak triangularizability. Given an $m \times m$ pattern E , construct an $(m+1) \times (m+1)$ pattern

$$E^+ = \{(i, j+1) : (i, j) \in E\}.$$

Thus E^+ is obtained by adding an all-zero first column and an all-zero last row.

Suppose first that E is weakly triangularizable. Use the weak triangularizing row order for the old rows, put the new zero row last, put the new zero column first, and then use the weak triangularizing column order for the old columns shifted one position to the right. Every old edge that formerly had position $i \leq j$ now has position $i < j+1$, so E^+ is strictly triangularizable.

Conversely, suppose E^+ is strictly triangularizable. In a strict triangularization of E^+ , move the zero column to the first column. This cannot destroy strictness, because nonzero columns only move weakly to the right. Next move the zero row to the last row. This cannot destroy strictness, because nonzero rows only move weakly upward. Now all old rows occupy row positions $1, \dots, m$ and all old columns occupy column positions $2, \dots, m+1$. For every old edge, strictness says

$$i < j+1.$$

Subtracting one from the old column positions gives $i \leq j$, which is a weak triangularization of E .

Thus E is weakly triangularizable if and only if E^+ is strictly triangularizable, proving NP-hardness. $\square$

Combining Theorem 6.2 with Theorem 4.2 gives an NP-complete sufficient condition for convexity. It does *not* prove NP-hardness of CONVEX, because the sufficient condition is not necessary, as shown by the diagonal and F examples.

7 A nonconvex pattern avoiding the simple certificates

The simple obstructions in Theorems 4.3 to 4.5 also do not characterize nonconvexity. The following small example is useful because it shows that even the “obstruction-avoidance” superset is strictly larger than the convex patterns.

Let

$$S_{\diamond} = \{(1, 2), (1, 3), (1, 4), (2, 1)\} \subseteq [4] \times [4].$$

For $X \in \text{SO}(4)$ write

$$y = (X_{12}, X_{13}, X_{14}) \in \mathbb{R}^3, \quad t = X_{21} \in \mathbb{R}.$$

Proposition 7.1. *Under the identification $\mathbb{R}^{S_{\diamond}} \cong \mathbb{R}^3 \times \mathbb{R}$ given by (y, t) ,*

$$P_{S_{\diamond}}(4) = \{(y, t) : \|y\|_2 \leq 1, |t| \leq \|y\|_2\}.$$

In particular, $P_{S_{\diamond}}(4)$ is nonconvex.

Proof. Let the first two rows of X be

$$u = (x, y) \in \mathbb{R} \times \mathbb{R}^3, \quad v = (t, w) \in \mathbb{R} \times \mathbb{R}^3.$$

Since u is a unit vector, $x^2 + \|y\|^2 = 1$. Since v is a unit vector in $u^{\perp}$, the value of its first coordinate is the value of the linear functional

$$v \mapsto \langle e_1, v \rangle$$

on the unit sphere of $u^{\perp}$. The norm of this functional on $u^{\perp}$ is

$$\|\text{proj}_{u^{\perp}} e_1\| = \sqrt{1 - \langle e_1, u \rangle^2} = \sqrt{1 - x^2} = \|y\|.$$

Therefore every point of $P_{S_{\diamond}}(4)$ satisfies $\|y\| \leq 1$ and $|t| \leq \|y\|$.

Conversely, suppose $\|y\| \leq 1$ and $|t| \leq \|y\|$. Choose x with $x^2 + \|y\|^2 = 1$ and put $u = (x, y)$. The preceding calculation shows that the functional $v \mapsto \langle e_1, v \rangle$ has norm $\|y\|$ on the three-dimensional space $u^{\perp}$. Hence its values on the unit sphere of $u^{\perp}$ contain the whole interval $[-\|y\|, \|y\|]$. We may choose a unit vector $v \in u^{\perp}$ with first coordinate t . Extending u, v to a positively oriented orthonormal basis of $\mathbb{R}^4$ gives a matrix in $\text{SO}(4)$ with the desired $S_{\diamond}$ -coordinates. This proves the displayed formula for $P_{S_{\diamond}}(4)$.

Now $(e_1, 1)$ and $(-e_1, 1)$ both lie in this set, where $e_1 = (1, 0, 0) \in \mathbb{R}^3$. Their midpoint is $(0, 1)$, which violates $|t| \leq \|y\|$. Hence $P_{S_{\diamond}}(4)$ is nonconvex. $\square$

Proposition 7.2. *The pattern $S_{\diamond}$ avoids the three simple obstruction families above: it contains no perfect matching plus extra edge, no pinned two-plane obstruction, and no rectangle $A \times B$ with $|A| + |B| > 4$.*

Proof. The pattern $S_{\diamond}$ uses only two rows, so it contains no perfect matching in $[4] \times [4]$. For the same reason, it cannot contain the pinned two-plane obstruction in dimension four, which uses three nonzero rows up to row and column permutations.

Finally, a full rectangle contained in $S_{\diamond}$ either uses only row 1, in which case it has size at most 1×3 and $|A| + |B| \leq 4$, or it uses row 2, in which case the columns common to rows 1 and 2 are empty. Thus no contained full rectangle satisfies $|A| + |B| > 4$. $\square$

Combining the examples above gives the strict containments

$$\begin{aligned} & \{S : S \text{ is strictly triangularizable up to transposition}\} \\ & \subsetneq \{S : P_S(n) \text{ is convex}\} \\ & \subsetneq \{S : S \text{ avoids the simple obstructions of Section 4}\}. \end{aligned}$$

The first strict containment is witnessed by D_n or F ; the second is witnessed by $S_\diamond$.

8 What is still missing for NP-hardness

The preceding results explain why the most direct reduction does not solve the problem. One has

$$S \text{ strictly triangularizable} \implies P_S(n) \text{ convex},$$

but the converse is false. Therefore NP-completeness of strict triangularizability does not imply NP-hardness of CONVEX.

A successful reduction from strict triangularizability would need gadgets that preserve convexity in the yes-instances while forcing genuine nonconvexity in the no-instances, without allowing the diagonal and F -type convex escape routes. One precise formulation is the following.

Conjecture 8.1 (A possible gadget separation lemma). *There is a polynomial-time map*

$$E \subseteq [m] \times [m] \mapsto S(E) \subseteq [N] \times [N], \quad N = \text{poly}(m),$$

such that:

- (i) *if E is strictly triangularizable, then $S(E)$ is strictly triangularizable up to transposition;*
- (ii) *if E is not strictly triangularizable, then $P_{S(E)}(N)$ is nonconvex.*

Proposition 8.2. *If Theorem 8.1 holds, then CONVEX is NP-hard.*

Proof. Reduce the NP-complete strict triangularizability problem of Theorem 6.2 to CONVEX. Given E , output $S(E)$. If E is strictly triangularizable, then $S(E)$ is strictly triangularizable up to transposition, so $P_{S(E)}(N)$ is convex by Theorem 4.2. If E is not strictly triangularizable, then $P_{S(E)}(N)$ is nonconvex by Theorem 8.1. Thus the reduction maps yes-instances to yes-instances and no-instances to no-instances. $\square$

Remark 8.3. The point of Theorem 8.1 is not that it is proved here; it is the missing statement. The diagonal, F , and $S_\diamond$ examples above are precisely why such a gadget, if it exists, has to do more than merely test non-strict-triangularizability.

9 Conclusion

This note obtains partial progress, not a solution of Ramachandran–Shu–Wang’s NP-hardness conjecture. The rigorous conclusions are:

- (1) CONVEX lies in $\forall\exists\mathbb{R}$ and hence in PSPACE under the dense support-matrix encoding.
- (2) Strict triangularizability is NP-complete.

- (3) Strict triangularizability is a sufficient but not necessary condition for convexity of $P_S(n)$.
- (4) Perfect-matching-plus-extra patterns, pinned two-plane patterns, and large rectangles give hereditary nonconvexity certificates.
- (5) These simple certificates do not characterize nonconvexity; the pattern $S_\diamond$ is a concrete counterexample.

The remaining open part is the genuine reduction from an NP-hard combinatorial property to nonconvexity of coordinate projections of $\text{SO}(n)$. The results above identify a natural route and the precise obstruction to completing it, but they do not by themselves prove NP-hardness of CONVEX.

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